DELHIVERY

Network Operations Intelligence

STRATEGY MEMO

1. The Scale of the Problem

Our graph-based audit of **26,369 trip-legs** across **1,657 facilities** and **2,783 corridors** confirms that the OSRM routing engine systematically underestimates delivery time. The numbers are stark:

98.2%	94.7%	2.06×	96%
of trips arrive later than OSRM predicts	of trips are chronically late (>20% over)	median actual time vs. OSRM estimate	of all 2,783 corridors are chronic

This is not a minor calibration issue. OSRM is structurally blind to facility dwell time, route-type constraints, and network cascading effects. The graph model built in this project addresses all three.

2. Root Cause: Dwell, Not Driving

The dominant source of delay is not road congestion — it is time spent *inside* facilities. Across the network, the median hub dwell time is **49 minutes** per parcel per stop. At the five worst hubs, that figure exceeds **2 hours**, meaning a parcel waits inside the building longer than it spends on the road.

Key Insight: Since the bottleneck is operational (dock capacity, sort throughput, dispatch scheduling) rather than infrastructural (roads, distance), the fix does not require building faster trucks or new highways. Targeted facility upgrades and smarter route-type choices deliver the ROI.

3. The Five Bottleneck Hubs Driving National Lateness

Graph-theory analysis — betweenness centrality, in/out-degree, and SLA-breach attribution — identifies five facilities responsible for **31.5%** of all delayed minutes nationwide.

Rank	Facility	City / State	Delay Share	Betweenness	Med. Dwell
1	**Bilaspur HB**	Gurgaon (Haryana)	14.7%	0.330	155 min
2	**Mankoli HB**	Bhiwandi (Maharashtra)	**6.3%**	0.104	142 min
3	**Nelmangala H**	Bangalore (Karnataka)	**5.5%**	0.187	140 min
4	Tathawde H	Pune (Maharashtra)	2.6%	0.065	112 min
5	Dankuni HB	Kolkata (West Bengal)	2.4%	0.136	116 min

The Gurgaon Chokepoint: Bilaspur HB alone accounts for **1 in every 7 delayed minutes** nationally. With a betweenness score of 0.330, one-third of all shortest-path routes in the network pass through it with near-zero redundancy. The top three hubs combined — Gurgaon, Bhiwandi, and Bangalore — drive **26.5%** of all national lateness.

4. Corridor-Specific Interventions

Three interventions address the highest-impact corridors with distinct cost profiles:

Intervention A — Gurgaon Facility Upgrade (High CapEx, Highest ROI)

- **Corridors:** Gurgaon → Bangalore running at 1.9× over OSRM, accumulating 89,000 excess minutes. Gurgaon → Kolkata at 2.0× and 56,000 excess minutes.
- **Root cause:** Morning dispatch backlog caused by dock saturation at Bilaspur HB (155-minute median dwell).
- **Recommendation:** Add dock doors and introduce a second sort wave to drain the AM backlog. Relieving this single node structurally reduces delay across all 49 outbound corridors simultaneously.

Intervention B — Bhiwandi Bypass Route (Low CapEx, Fast ROI)

- **Root cause:** Local Mumbai-bound volume is unnecessarily routed through the Mankoli HB sort facility, adding 142 minutes of median dwell.
- **Recommendation:** Establish a direct line-haul bypassing the Bhiwandi sort for Mumbai-bound parcels. This is a routing logic change, not a construction project — minimal capital expenditure, fast execution.

Intervention C — Bangalore FTL Shift (Zero CapEx, Immediate)

- **Root cause:** Carting consolidation wait drives the bulk of the 140-minute median dwell at Nelmangala H on short, high-frequency outbound lanes.
- **Recommendation:** Shift to FTL on ∼**200 flagged corridors** network-wide where FTL saves 2.5–4 hours per shipment. Carting remains the cheaper default everywhere else.

5. Graph-Enhanced ETA Model: Proven Improvement

Three model tiers were built and benchmarked, each adding a layer of spatial network awareness. Results on the held-out test set:

Model	MAE (min)	Within 10%	Within 15%	Within 20%	Best?
Baseline (Tabular)	50.2	–	28.0%	35.9%	
Node2Vec Enhanced	37.0	–	37.7%	46.8%	
GraphSAGE (GNN)	37.4	–	37.6%	47.2%	

The graph-enhanced models reduce MAE by **26%** over the tabular baseline — from 50.2 to ∼37.2 minutes. SLA compliance at the 15% accuracy threshold jumps by 35% (28.0% → 37.6%). GraphSAGE edges out Node2Vec on the 20% business metric (47.2% vs 46.8%) by aggregating live congestion features from downstream hubs, not just spatial position.

Business Translation: Deploying GraphSAGE to production today — before any physical hub upgrade — immediately sets more accurate customer ETAs, reducing SLA breach complaints on the software side while physical fixes are executed.

6. Projected Impact & ROI

Upgrading the top three hubs (targeting a realistic 40% reduction in dock-and-sort dwell time) combined with the Bhiwandi bypass and Bangalore FTL shift delivers:

- **26.5%** of total national delayed minutes addressed by the top-3 hub interventions alone.
- ∼**200 corridors** flagged for FTL conversion, each saving 2.5–4 hours per shipment with zero construction required.
- **Late-delivery rate** on downstream lanes of top hubs projected to fall from ∼89% toward ∼75% post-upgrade.
- Bhiwandi bypass and Bangalore FTL shift represent the **highest immediate ROI** due to near-zero CapEx — both are routing and scheduling changes only.

Note: Financial projections should be calibrated with the Finance team using actual per-delivery cost rates. Model inputs (INR 32/km FTL, INR 10/km Carting) are illustrative baselines built into the framework and can be updated directly.

7. Final Recommendation — Prioritised Action Plan

#	Action	Timeline	Expected Outcome
1	Deploy GraphSAGE ETA model to production	Immediate	26% MAE reduction; honest ETAs while hubs are upgraded
2	Launch Bhiwandi bypass route (direct line-haul)	2–4 weeks	Eliminates sort-facility wait on Mumbai-bound volume
3	Activate FTL on 200 flagged corridors	2–4 weeks	2.5–4 hrs saved per shipment, zero CapEx
4	Gurgaon dock expansion + 2nd sort wave	3–6 months	Relieves 14.7% of national delay; unblocks 49 downstream lanes
5	Bangalore & Kolkata hub capacity review	6–12 months	Addresses remaining 7.9% of top-5 combined delay share